Assignment for the Course "Current Research Areas II " 2026

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What was the plan?

- Develop a structure for the lawn mower operations so it would:
  - Stay in the designated area
    - Prepare the base system following the logic of separate turn states
  - Cover the area as efficient as possible(with in our ability)
    - With the base system working find the most efficient angle
    - Analyse and find 3 possible improvements based on the results
      - Implement them and note which provided the biggest coverage increase
  - Stop if outside of the area for more than 3 sec.
    - Prepare the safety state.

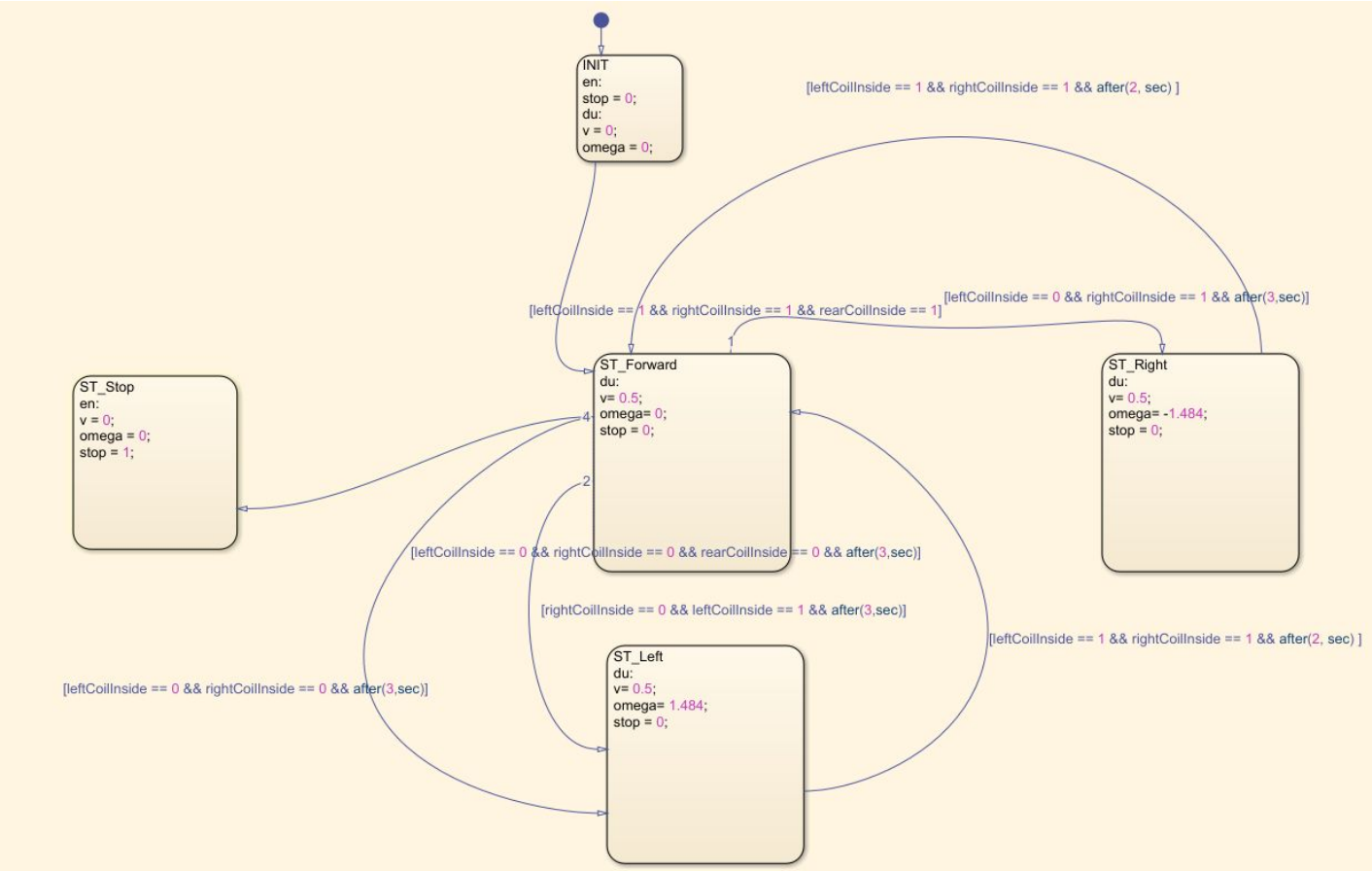
Plan execution

- **Stay in the designated area**
  - Initially we failed to do so, as the mower would breach the area line and move indefinitely or stop per the condition.
  - The main cause of this issue was not found.
  - From the visualisation and action plots it was noticed that it would fail to continue moving backwards and so the approach was slightly changed.
  - Now the object when reaching limits turns around(makes a U turn).
    - In case both coils are outside, we move just left(could be right as well). If just one is outside a wide turn is made(using delay and omega)
    - This approach finally give us positive results

Charts & results

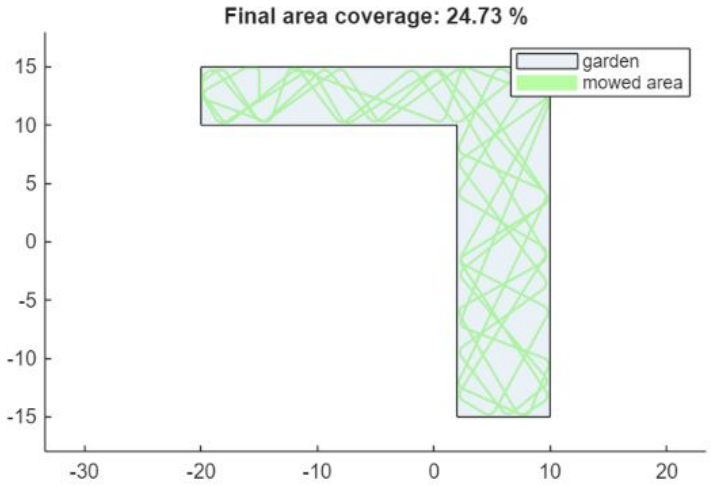
We set it up to have an angle of ~170deg. Which is achieved with time of 2sec. and omega of 1.484.

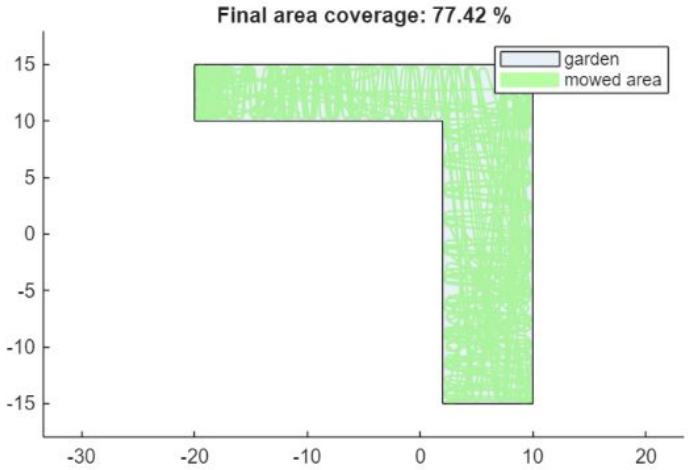
$$\Delta\theta(\text{degrees}) = (\omega \times d) \times \frac{180}{\pi}$$



Charts & results

Here is a visible comparison of the covered area when the rotation angle is 90deg and 170deg.

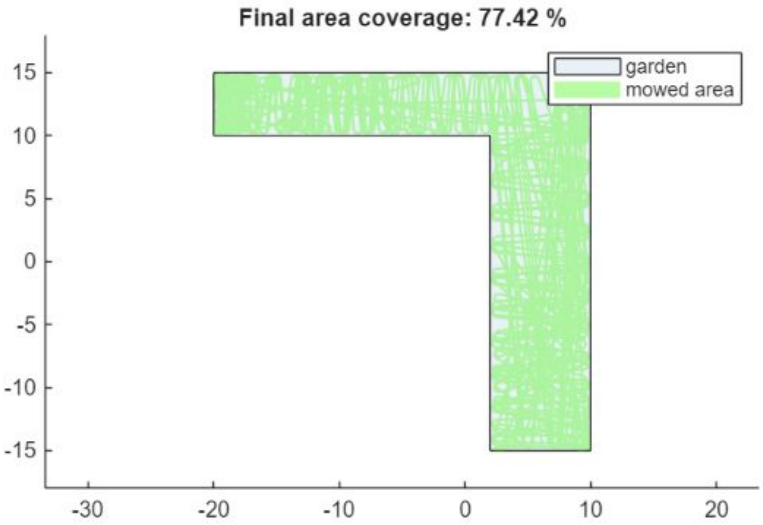




Plan execution

- **Cover the area as efficient as possible**
  - The successful method of the u-turns still is considered **random** and with angle tuning we are able to cover the area of 77.42%

Why 170deg:
Its an imperfect angle from the full turn of 180. Each pass is slightly offset from the last, ensuring that overlap is kept to a minimum, without additional algorithms.



The Algorithm future improvements

- **Narrow corner escape**
  - Based on the visualisation it can be observed that a lot of time wasted and overall lap is done in the sharp corners as the system continues to sharply turn creating a “ping-pong” effect. To resolve this a memory of a bounce could be introduced and in case the “wall” is hit multiple times within ~1sec the angle for the turns is changed to a wider angle, helping the mower escape.
- **Forced area splitting**
  - Per the visualization we can the that the “long” paths are not as efficient. To change this a forced u-turn can be introduced in case no “wall” is hit in ~5sec. with an additional cool-down setting, allowing to escape this section after its completed.
- **Boundary covering**
  - Ensure that the first move is to follow the boundary. The mower deliberately tries to keep the boundary wire directly between the left and right coils. This will significantly increase the covered area, by a relatively straight forward initial move.

The progress

- To much time was spent on understanding and setting the system up to even cover the area at all.
- There is a direct path for improvements that should be relatively easy to implement without additional “sensors” on the system and complicated memory/path saving.

Day 2- Agenda

- Prepare a live script which is connected with our model
- Add interactive elements
  - Omega for right and left turns
  - Time delay for the turn
- Prepare 3 different mower silhouette
  - Existing model
  - Narrow model
  - Wide model
- Prepare 3 different lawn areas
  - Get inspiration from AI
    - Circle, square and L-shape
- Compare the results

Prepare a live script which is connected with our model

- With our additional features we decided to abandon the existing live script and start on our own, which does support us and gives us extra features for the coverage.

```
--- Final Results ---
Model: tasks2
Total Sim Time: 6000.00 s
Garden Area: 330.00 m²
Total Mowed Area: 269.43 m²
Final Coverage: 81.65 %
```

Add interactive elements

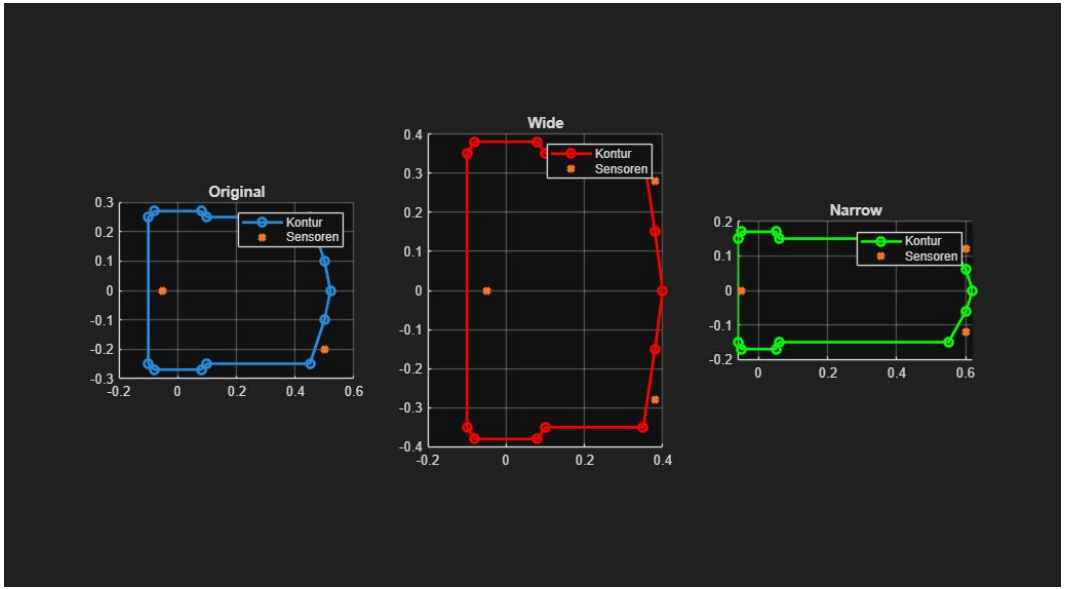
- Turning time delay and turning angle has been made into a scaler giving the opportunity to change it on the go. Showing how the angle impacts the results without the need to go into the simulation it self.
- We were not able to add the garden shape and mower shape as an automated button due to time constraints.

```
%% 1. Interactive Parameters
% Adjust these values using Live Editor sliders
target_omega = 1.484 ;  % Angular velocity (rad/s)
target_time  = 2 ; % Turn duration (seconds)
turn_angle=(target_omega*target_time)*57.29;
fprintf('Turn angle: %s\n', turn_angle);
stop_time = 6000;  % Total simulation time (seconds)
```

```
Turn angle: 1.700367e+02
```

Prepare 3 different mower silhouette

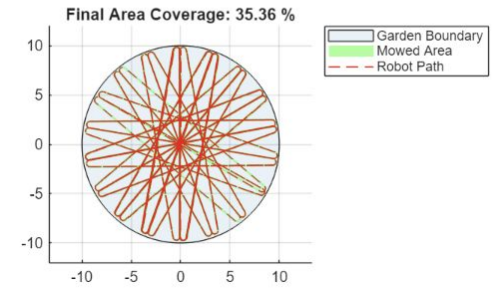
- The silhouette and most importantly their coil placement has been changed giving us different results. We went for 3 models: original, narrow and wide.



Prepare 3 different lawn areas

- Three different garden areas have been prepared. A circle, original L-shape and a square.
- AI was used to generate the shape.
- Its clearly seen how the corners and lack there of changes the existing models ability to cover the area.
- We did not find the time to properly automate the different shape process.

```
--- Final Results ---
Model: tasks2
Total Sim Time: 6000.00 s
Garden Area: 314.16 m²
Total Mowed Area: 111.09 m²
Final Coverage: 35.36 %
```



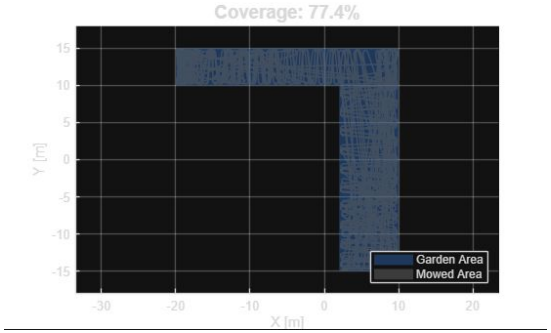
The comparison

The angle and the shapes:

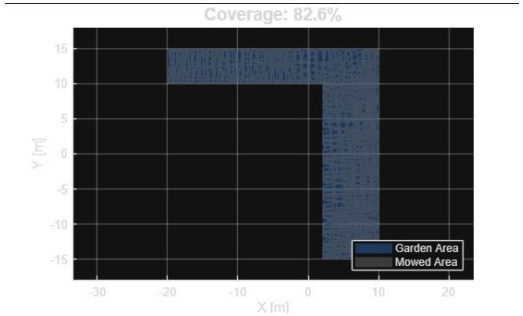
- **The L-shape** angle stands the same as before with 170deg being the perfect imperfect angle that makes sure the turn is narrow enough to not spend a lot of time wondering around and small enough not to overall and trap itself.
- **The Square** benefits for a bit of a wider angle that will break even more. But most notably the wide free area where the mower just crosses wastes a lot of time, is not precise and tends to overlap.
- **The Circle** especially as the starting point is 0,0 is a difficult shape, but the 170 angle is optimal as it escapes any traps in a small square(as it would for 90deg) and reaches the outer layers, but it still tends to overlap.

Comparison of the 3 silhouettes

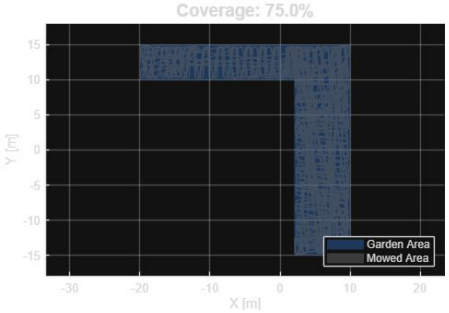
- We tested those 3 silhouettes on the L-shape garden and we got these results :



Original (77.4%)



Wide (82.6%)



Narrow (75%)

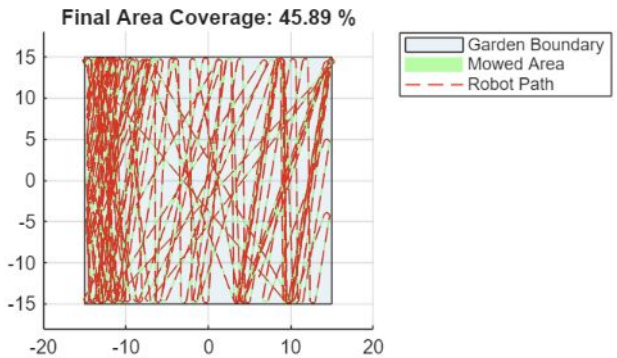
The wide-bodied robot showed an improvement of 5.2 percentage points compared to our latest best results (original body, angle: 170°).

Improvement for the future : changed the velocity and/or the angle, create a wider body ?

The Coperison: The Square under different angle and mower

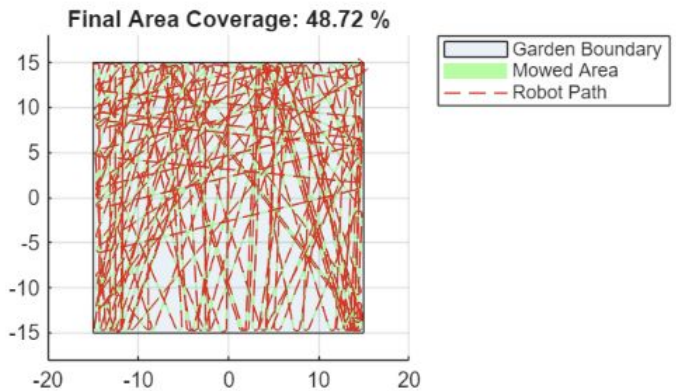
Original shape with 170deg

```
--- Final Results ---
Model: tasks2
Total Sim Time: 6000.00 s
Garden Area: 900.00 m²
Total Mowed Area: 412.99 m²
Final Coverage: 45.89 %
```



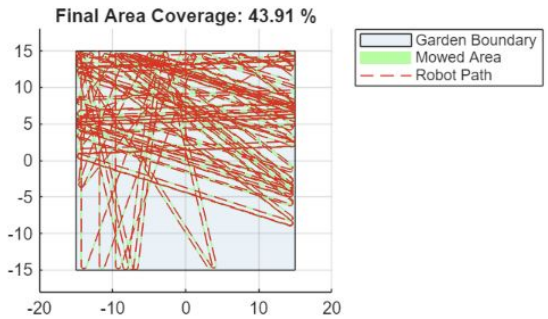
Original shape with 160deg

```
--- Final Results ---
Model: tasks2
Total Sim Time: 6000.00 s
Garden Area: 900.00 m²
Total Mowed Area: 438.52 m²
Final Coverage: 48.72 %
```



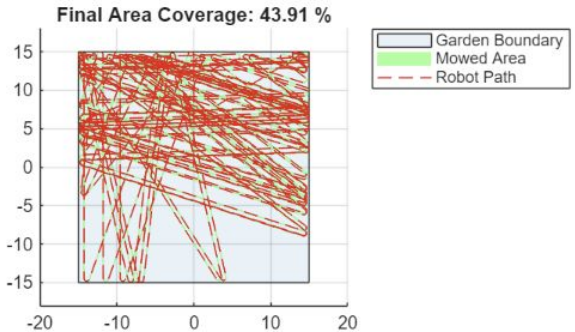
Wide mower

```
--- Final Results ---
Model: tasks2
Total Sim Time: 6000.00 s
Garden Area: 900.00 m²
Total Mowed Area: 395.20 m²
Final Coverage: 43.91 %
```



Narrow mower

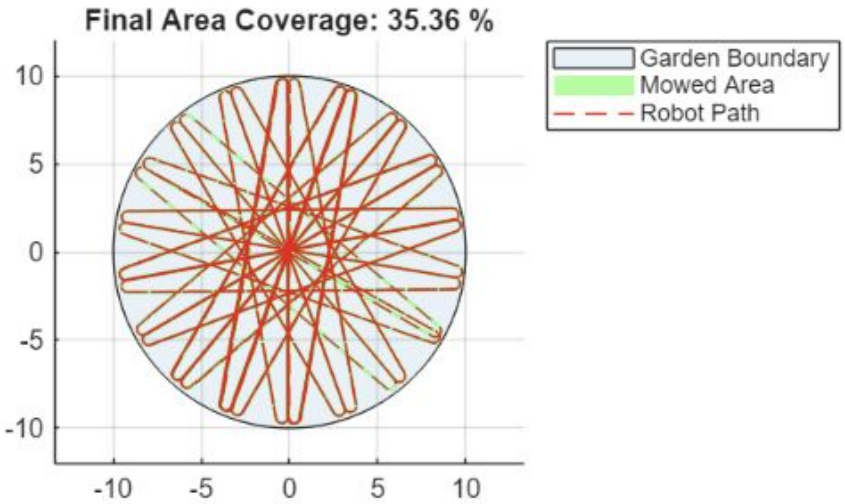
```
--- Final Results ---
Model: tasks2
Total Sim Time: 6000.00 s
Garden Area: 900.00 m²
Total Mowed Area: 395.20 m²
Final Coverage: 43.91 %
```



Circle

The simulation took too much time to simulatie so no extra graphs were produced.

```
--- Final Results ---
Model: tasks2
Total Sim Time: 6000.00 s
Garden Area: 314.16 m²
Total Mowed Area: 111.09 m²
Final Coverage: 35.36 %
```



The use of AI

We are using it somewhere in between vibe coding and developer, as we are aware of what is done, what is to be implemented, some changes are done in the code by hand, but most work is done by AI.